Budde Dhathvi Charan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.32
Narayana Junior College	Class XII (Telangana Board of Intermediate Education)	2024	97.80%
Jawahar Navodaya Vidyalaya Nizam Sagar Nizamabad Telangana	Class X (Central Board of Secondary Education)	2022	92.00%

RELEVANT COURSES

- **ComputerScience:** Algorithms, Data Structures, Operating Systems, Database Management Systems, Computer Architecture, Theory of Computation, Discrete Maths, Software Development Fundamentals, Programming in C.
- **Mathematics:** Linear Algebra, Probability and Random Variables, Statistics, Calculus, Transform Techniques.

PROJECTS

xv6 Kernel Development

- Redesigned xv6's scheduler by implementing a 4-level System-Call-Aware MLFQ with dynamic priority adjustment, starvation prevention, and per-process scheduling statistics **for workload-aware process scheduling.**
- Extended xv6's virtual memory subsystem with Clock-based page replacement, scheduler-aware eviction, swap management, and per-process memory statistics to **improve memory utilization** under constrained physical memory
- Implemented disk-backed swap with configurable FCFS/SSTF disk scheduling and RAID 0/1/5 storage management supporting parity-based fault recovery.

Arbitrary Precision Arithmetic Library

- Implemented an arbitrary-precision arithmetic library in Java using **object-oriented programming principles**.
- Built custom **AInteger** and **AFloat** classes supporting arithmetic operations on arbitrarily large integers and floating-point numbers, enabling precise computation beyond native data type limits.
- Packaged the library as a reusable JAR, automated builds using Apache Ant, and developed a Python CLI for execution.

Real-Time Server Fleet Monitoring Platform

- Built a Node.js, React, and PostgreSQL dashboard to visualize live CPU/RAM metrics across multiple servers.
- Replaced HTTP polling with WebSockets (socket.io) to stream instant data updates to dynamic React charts.
- Optimized PostgreSQL query performance by implementing timestamp indexing for high-speed data retrieval, and configured an automated background cron job to continuously prune stale server metrics to prevent storage overload.

Integer Workload Optimization using Floating-Point Units

- Profiled integer-heavy workloads on GEM5(RISC-V), Intel i9, and ARM Cortex-A76 to identify performance bottlenecks.
- Optimized compute-intensive applications including **matrix multiplication, 2D convolution, and determinant computation** using loop tiling, loop reordering, and floating-point offloading to improve instruction-level parallelism.
- Implemented integer, floating-point, and hybrid variants of compute-intensive workloads to evaluate architecture-aware optimization strategies across simulated and real hardware.

WORK EXPERIENCE

Research Intern | Department of CSE, IIT Bombay

- Assisted in developing educational material for a comprehensive **Machine Learning Book**, covering mathematical foundations, algorithms, implementation, and deep learning concepts.
- Contributed to implementation examples, technical content preparation, and validation of machine learning concepts under faculty supervision

SKILLS

Languages: C++, C, Python, Java, SQL, HTML/CSS, JavaScript, RISC - V Assembly, Shell Script

Frameworks/Tools: Git, Github, CMake, Node.js, Express.js, PostgreSQL, Docker

Soft Skills: Communication, Leadership, TeamWork

Competitive Programming: Codeforces

POSITIONS OF RESPONSIBILITY

Undergraduate Representative | Student Gymkhana Council | **May 2026 - Present**

- Represent undergraduate students in the Student Gymkhana Council and communicate student concerns to the administration.
- Coordinate with council members to organize student initiatives and improve campus engagement.

Head, Education Program Management Domain | Prayas | **2025-26**

- Led the Education Program Management domain, overseeing planning and execution of educational initiatives.

Hospitality Coordinator | Milan | **2025-26**

EXTRACURRICULAR

- **NSO -** Table Tennis